

Physics Theory Compendium

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1 Orientation

This note organizes physics around a common template: choose the fields, symmetries, and action, derive equations from stationarity, identify conserved currents through symmetry, and pass to Hamiltonian or quantum formalisms when needed. The point is not to list isolated formulas, but to show how classical mechanics, field theory, relativity, gauge theory, and continuum models fit inside one variational language.

2 Universal Variational Formulation

2.1 Fields and Action

The starting object is a collection of fields

$$u^\alpha(x), \quad x = (x^0, x^1, \dots, x^d),$$

defined over spacetime or over a configuration domain. Their dynamics are encoded by a Lagrangian density

$$\mathcal{L} = \mathcal{L}(x, u, u_J),$$

where u_J denotes derivatives of the fields. The action is

$$S[u] = \int_{\Omega} \mathcal{L}(x, u, u_J) dx.$$

2.2 Stationary Action and Field Equations

Physical equations appear by requiring the action to be stationary:

$$\boxed{\delta S = 0.}$$

For local theories this yields the Euler–Lagrange system

$$\boxed{\sum_J (-D)_J \frac{\partial \mathcal{L}}{\partial u_J^\alpha} = 0,}$$

where D_J is the total derivative associated with the multi-index J . This single schema includes Newtonian mechanics, Maxwell theory, Yang–Mills theory, general relativity, and continuum mechanics.

2.3 Canonical Momenta and Hamiltonian Density

When time is separated from space, the canonical momentum is

$$\pi_\alpha = \frac{\partial \mathcal{L}}{\partial(\partial_t u^\alpha)}.$$

The Hamiltonian density and Hamiltonian are then

$$\mathcal{H} = \pi_\alpha \partial_t u^\alpha - \mathcal{L}, \quad H = \int \mathcal{H} dx.$$

This reframes dynamics as evolution on phase space and becomes the bridge to canonical quantization and statistical mechanics.

2.4 Noether's Theorem

Continuous symmetries generate conserved currents:

$$\partial_\mu J^\mu = 0.$$

The corresponding conserved quantity is

$$Q = \int J^0 d^3x.$$

Time translation gives energy, spatial translations give momentum, rotations give angular momentum, and gauge symmetry constrains admissible interactions.

3 Quantum and Statistical Formulations

3.1 Generating Functionals

The quantum theory of a field is encoded by the generating functional

$$Z[J] = \int \mathcal{D}u \exp \left(iS[u] + i \int J_\alpha u^\alpha dx \right).$$

From it one obtains the connected functional

$$W[J] = -i \log Z[J],$$

the effective action

$$\Gamma[u] = W[J] - \int J_\alpha u^\alpha dx, \quad u^\alpha = \frac{\delta W}{\delta J_\alpha},$$

and correlation functions

$$\langle u(x_1) \cdots u(x_n) \rangle = \frac{1}{Z} \frac{\delta^n Z}{i^n \delta J(x_1) \cdots \delta J(x_n)} \bigg|_{J=0}.$$

3.2 Imaginary Time and Statistical Mechanics

After Wick rotation, oscillatory amplitudes become exponential weights. This connects quantum field theory with equilibrium statistical mechanics and shows why the Hamiltonian controls both time evolution and thermodynamic ensembles.

4 Classical Mechanics

4.1 Finite-Dimensional Configuration Spaces

For generalized coordinates $q(t)$, the action is

$$S[q] = \int L(q, \dot{q}, t) dt, \quad L = T - V.$$

The Euler–Lagrange equation becomes

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = 0.$$

This is the prototype for all later field equations: kinetic structure minus potential or interaction structure, varied against admissible trajectories.

4.2 Hamilton's Equations

With canonical momentum

$$p = \frac{\partial L}{\partial \dot{q}},$$

the Hamiltonian is

$$H(q, p, t) = p\dot{q} - L,$$

and motion is rewritten as

$$\dot{q} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial q}.$$

This makes symplectic structure explicit and prepares the route to both canonical quantization and transport theory.

5 Relativity

5.1 Special Relativity

The invariant interval is

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu.$$

For a free relativistic particle,

$$S = -m \int ds.$$

Lorentz invariance forces space and time to be treated as parts of a unified geometric object, and field equations must respect this invariance.

5.2 General Relativity

The gravitational field is the metric $g_{\mu\nu}$. Its action is the Einstein–Hilbert functional

$$S = \frac{1}{16\pi G} \int R \sqrt{-g} d^4x.$$

Equivalently, the Lagrangian density is

$$\mathcal{L} = \frac{R}{16\pi G} \sqrt{-g}.$$

Variation with respect to the metric gives Einstein's equations

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

Here geometry is dynamical: curvature is not a background ingredient but the field itself.

6 Classical Field Theories

6.1 Electromagnetism

The electromagnetic potential is A_μ , with field tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

The Lagrangian density is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - J^\mu A_\mu.$$

Variation gives Maxwell's equations with sources, while the antisymmetry of $F_{\mu\nu}$ produces the Bianchi identity automatically.

6.2 Scalar Fields

For a real Klein–Gordon field ϕ ,

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2.$$

The resulting equation,

$$(\square + m^2)\phi = 0,$$

is the relativistic prototype for linear wave propagation with mass.

6.3 Spinor Fields

For a Dirac spinor ψ ,

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi.$$

This first-order Lagrangian captures relativistic fermions and makes spin and antiparticle structure intrinsic rather than added by hand.

6.4 Nonrelativistic Quantum Fields

For a Schrödinger field $\psi(\mathbf{r}, t)$,

$$\mathcal{L} = i\psi^* \partial_t \psi - \frac{1}{2m} |\nabla \psi|^2 - V|\psi|^2.$$

This is the field-theoretic form of the Schrödinger equation and a natural entry point for many-body theory and condensed matter models.

7 Gauge Theories

7.1 Quantum Electrodynamics

For fields (ψ, A_μ) , the covariant derivative is

$$D_\mu = \partial_\mu + ieA_\mu,$$

and the QED Lagrangian is

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}.$$

Gauge symmetry determines the interaction term and ties charge conservation to local phase invariance.

7.2 Yang–Mills Theory

For a nonabelian gauge field A_μ^a ,

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc} A_\mu^b A_\nu^c.$$

Its pure gauge Lagrangian is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu}.$$

The extra commutator term encodes gauge self-interaction, which is absent in electromagnetism.

7.3 Quantum Chromodynamics and the Standard Model

QCD uses the gauge group $SU(3)$ with quark and gluon fields:

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi - \frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu}.$$

The Standard Model unifies the gauge sectors through

$$SU(3) \times SU(2) \times U(1),$$

with total Lagrangian

$$\mathcal{L} = \mathcal{L}_{\text{Gauge}} + \mathcal{L}_{\text{Fermion}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}}.$$

This is a clean example of the general philosophy: specify fields, symmetry, and local invariant terms, then derive all equations and conserved quantities.

8 Continuum Physics

8.1 Continuum Mechanics

For a displacement field $u(x)$, the deformation gradient is

$$F = I + \nabla u.$$

A representative Lagrangian density is

$$\mathcal{L} = T - W(F), \quad T = \frac{1}{2}\rho \dot{u}^2.$$

This packages elasticity and nonlinear material response into the same action framework used in particle and field theory.

8.2 Fluid Mechanics

For density ρ and velocity $\mathbf{v}$,

$$\mathcal{L} = \frac{1}{2}\rho v^2 - \varepsilon(\rho).$$

With suitable constraints or auxiliary potentials, this variational structure recovers continuity and momentum balance equations.

8.3 Linear Elasticity

The strain tensor is

$$\varepsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i),$$

and Hooke's law reads

$$\sigma_{ij} = C_{ijkl}\varepsilon_{kl}.$$

This is the small-deformation limit of the broader continuum framework above.

9 Extended Theories

9.1 String Theory

For a worldsheet embedding $X^\mu(\sigma, \tau)$, the Polyakov action is

$$S = -\frac{T}{2} \int d^2\sigma \sqrt{-h} h^{ab} \partial_a X^\mu \partial_b X_\mu.$$

The dynamical variable is no longer a point trajectory or spacetime field, but a map from a two-dimensional worldsheet into spacetime.

9.2 Effective Viewpoint

Even when a theory is not fundamental, the variational language still applies. Effective theories collect all symmetry-compatible operators at a chosen scale, with the dominant terms defining the leading physics and higher-order terms recording controlled corrections.

10 Unified View

Across these examples, a physical theory is characterized by

Configuration space,
fields u^α ,
symmetry group,
Lagrangian density $\mathcal{L}$,
action S ,
 $\delta S = 0$,
Euler–Lagrange equations,
Noether currents,
Hamiltonian H ,
 $Z[J]$,
correlation functions.

That common structure is what makes it possible to read mechanics, field theory, gravity, and continuum models as different realizations of one mathematical engine rather than disconnected subjects.